

A 3D architectural rendering of a residential neighborhood. The central focus is a two-story house with a dark blue roof and light-colored walls. The house is cutaway, revealing its internal structure and energy systems. On the roof, there are several blue solar panels. Inside, a green network of pipes and ducts is visible, connecting various components like a water heater, air conditioning units, and a boiler. The interior shows a living room with a sofa, a bedroom with a bed, and a kitchen. The house is surrounded by other similar houses in the background, and a red car is parked on the street in the foreground.

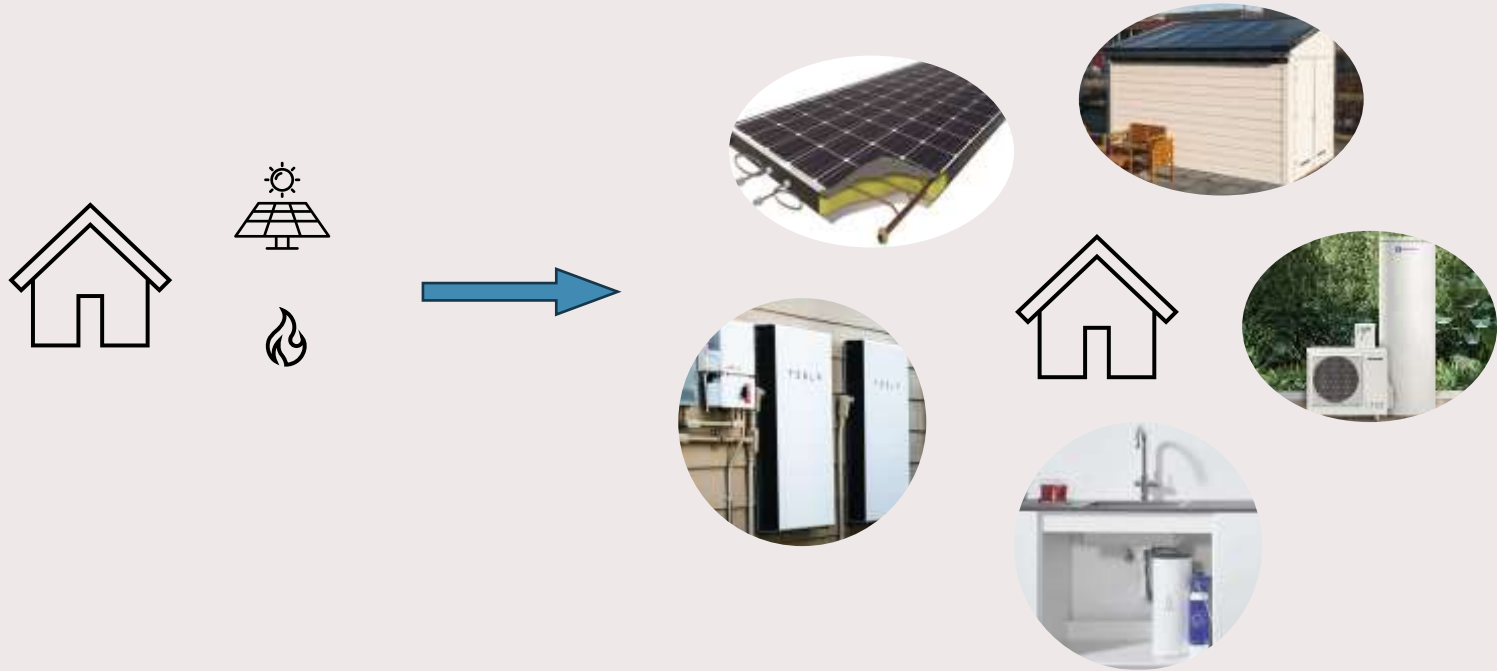
The impact of control strategies on design decision of residential energy systems

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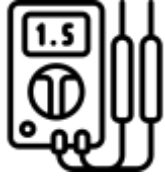
Promoter: dr.ir. Roel Loonen

Context

There are many new technologies that have come in the market to increase the renewable energy implementation in the building sector, and transition of energy systems from gas to electricity



Context



Empirical, isolated
studies



Building simulation
studies with simple
controls



Control
optimization for a
specific design

Research objective

When new technologies are evaluated with simple controls, they do not get a fair representation and hence could lead to biased judgement when it comes to decision making

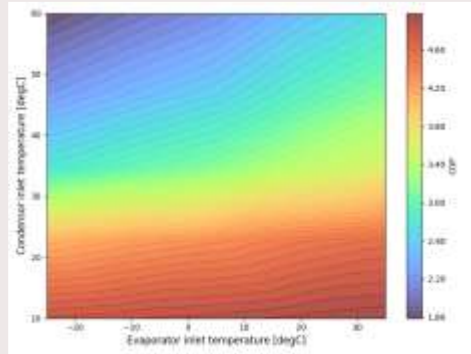
Objective:

To evaluate the impact of design decisions in selecting building energy systems when appropriate control strategies are taken into consideration

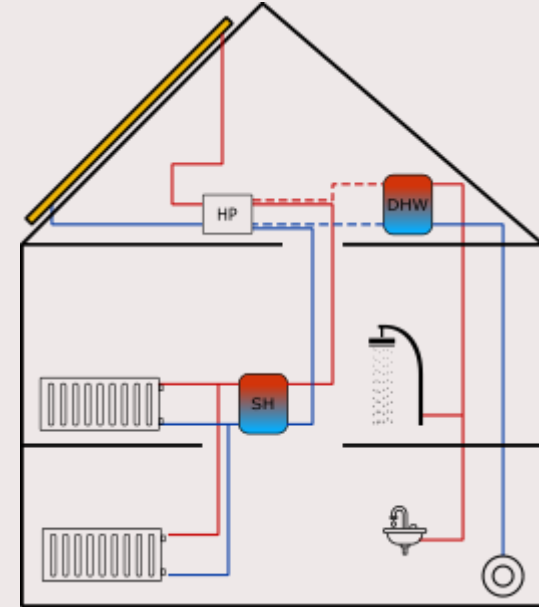
Case study

Simple terraced house with PVT-HP system

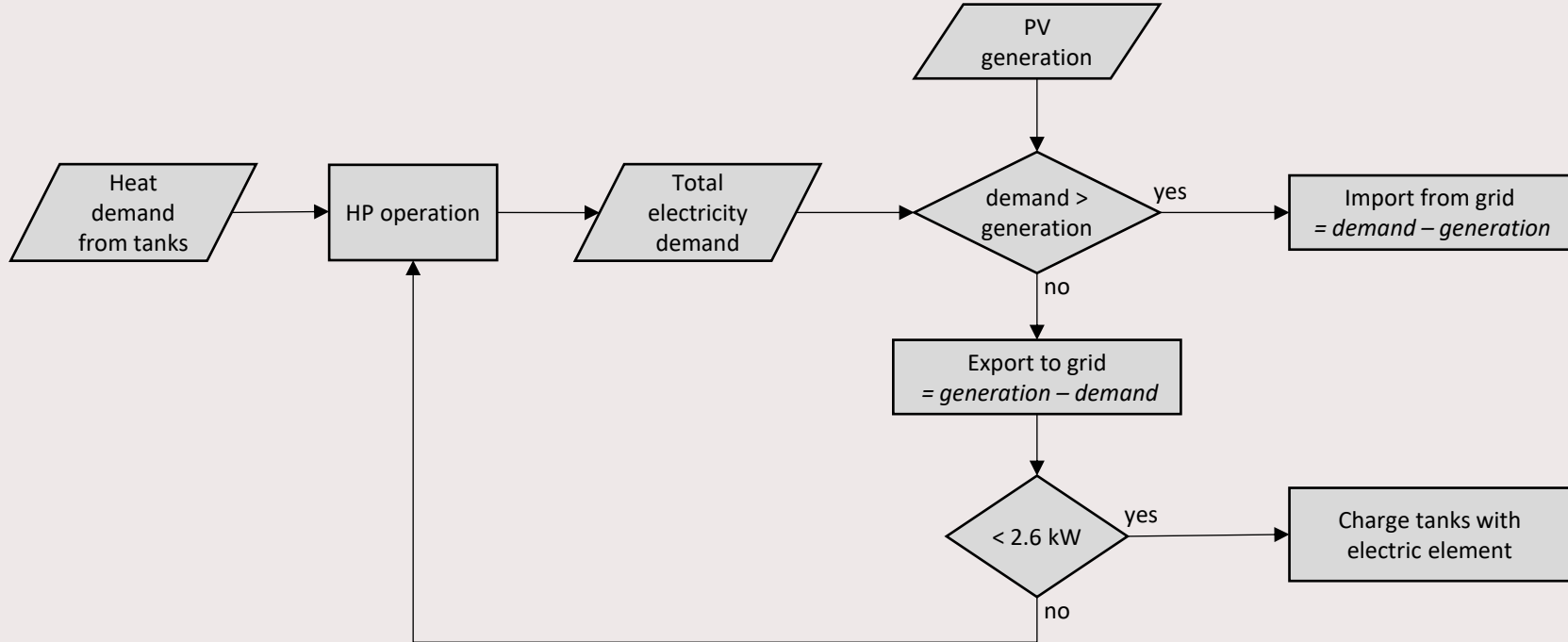
- Brine-water heat pump
 - can operate at source side temperatures of $[-25, 30]^{\circ}\text{C}$
 - Performance of the heat pump is based on entering source and load side fluid temperatures



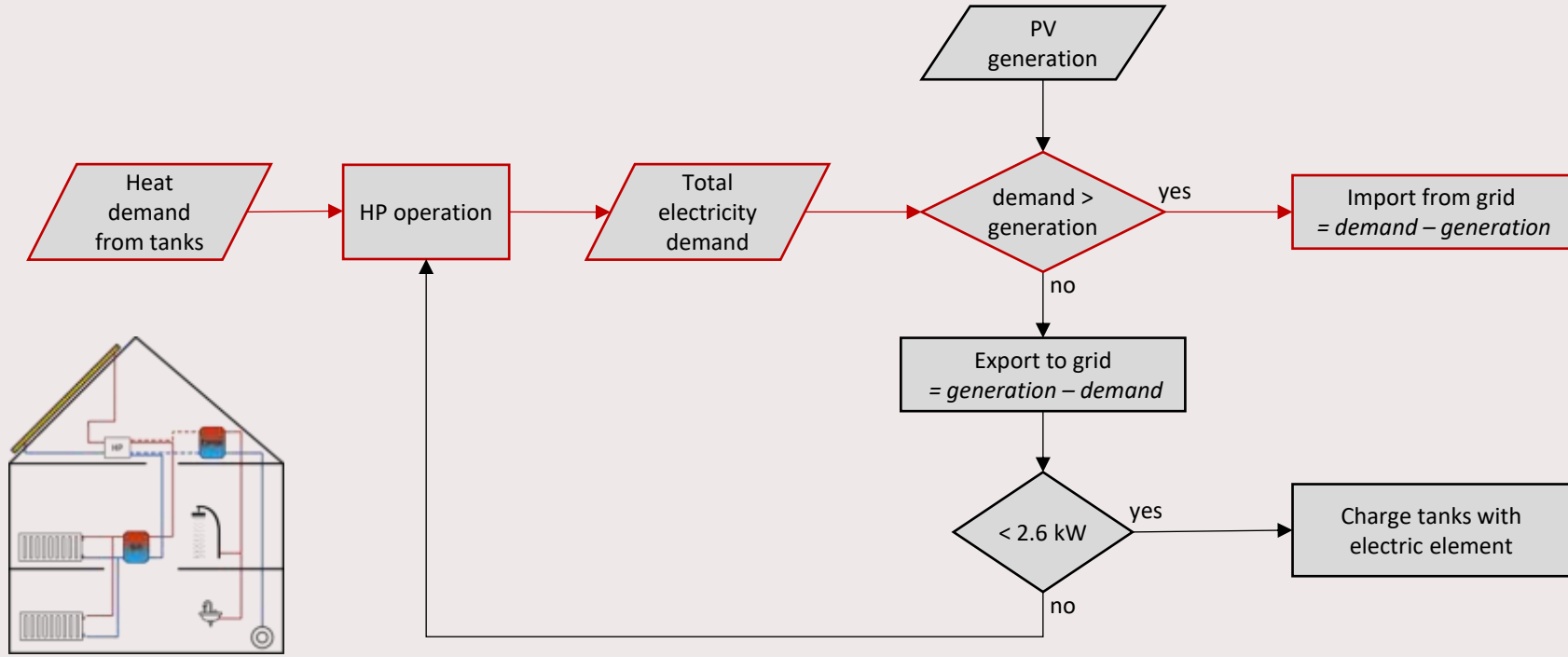
- Thermal storage in DHW tank and SH buffer tank
 - Define the demand for the heat pump



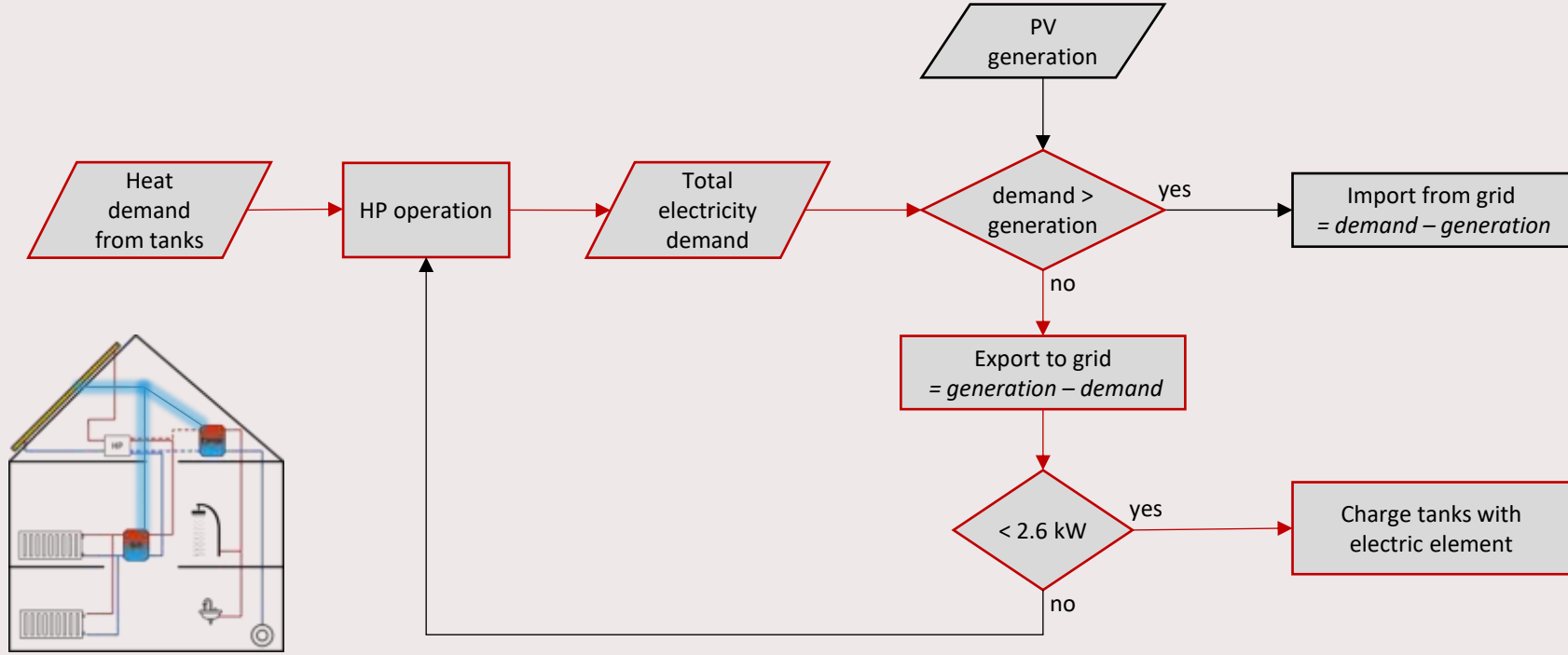
Control strategy definition



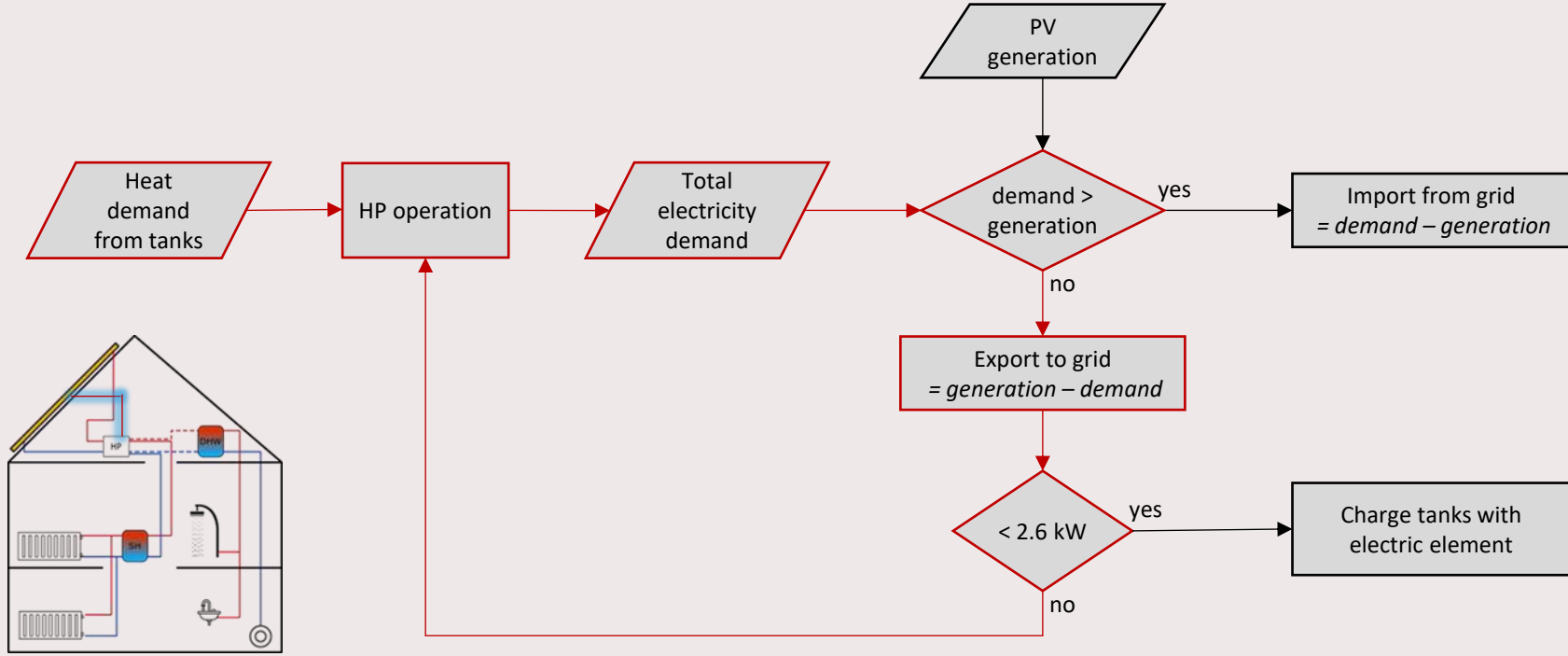
Control strategy definition



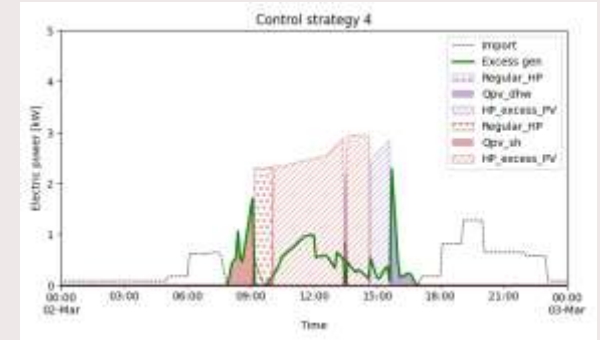
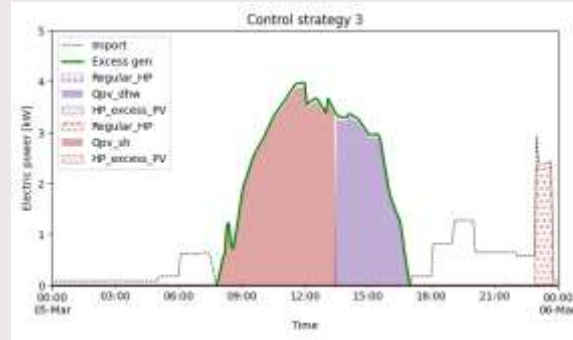
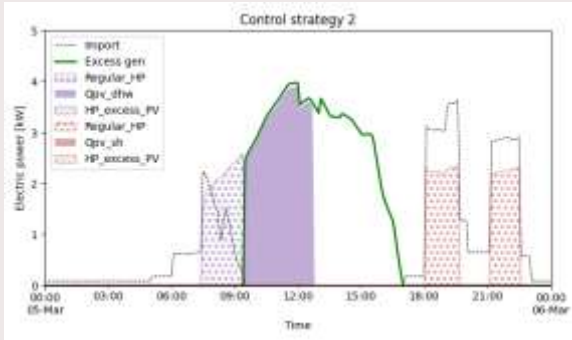
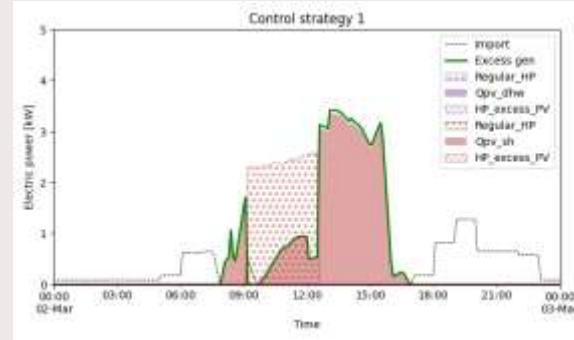
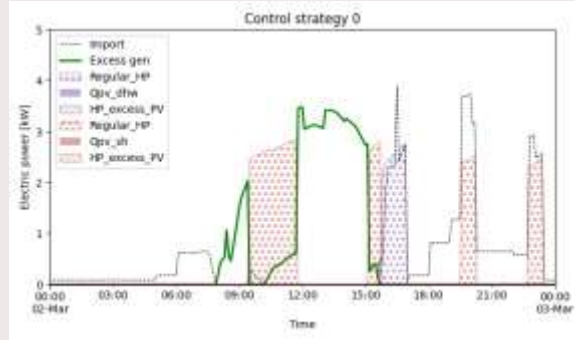
Control strategy definition



Control strategy definition



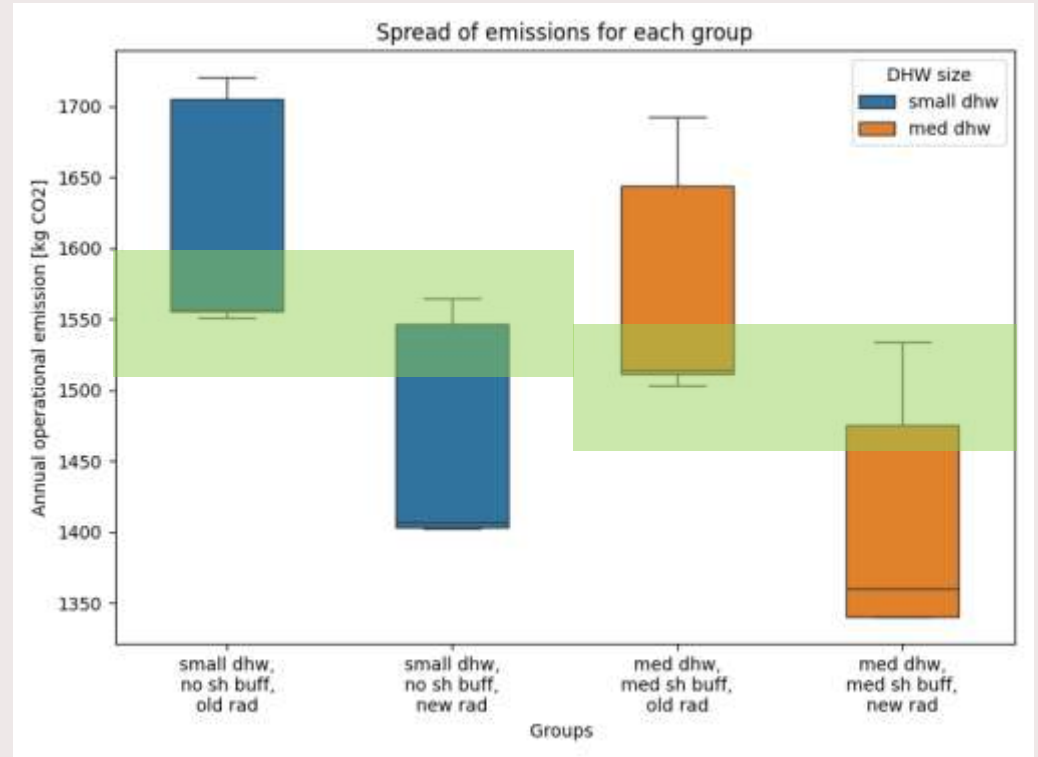
Demonstration of control strategies



Results 1

For 2 groups of tanks sizes → comparing between a system with old radiator and a new radiators

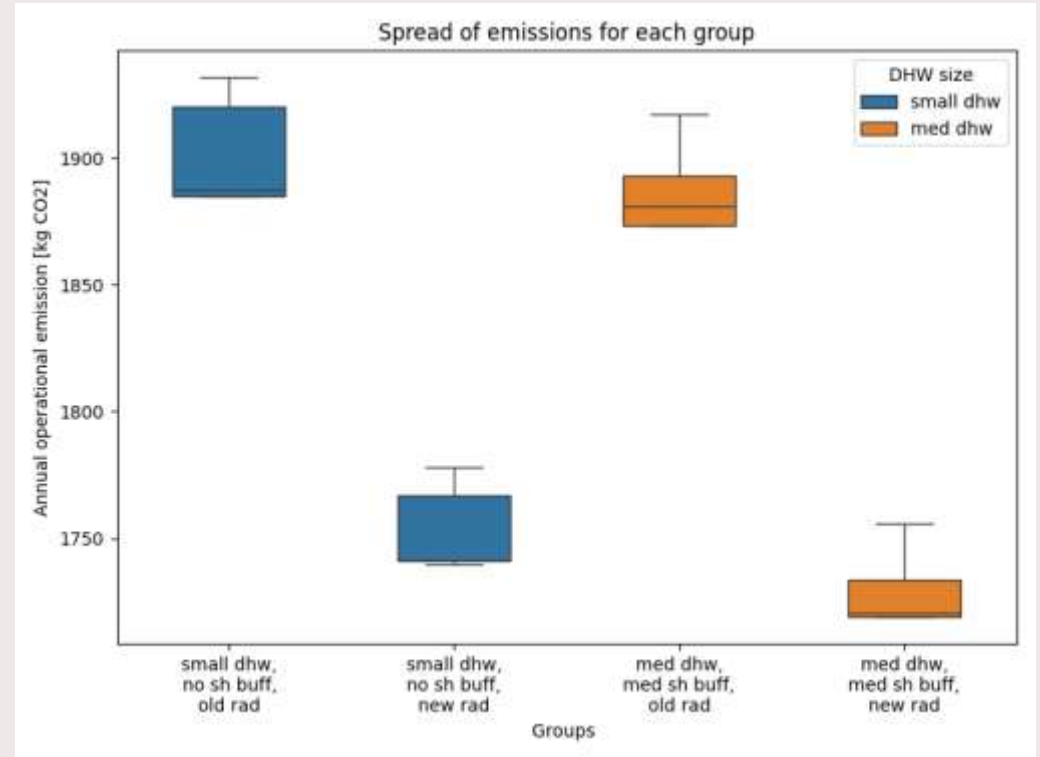
- Broader performance range when control strategies are considered
- Base control strategies show that changing radiators can be beneficial; however the right controls can lead to similar performance
- Bigger the storage tank, higher is the potential for operational emission reduction



Results 1

Same tank combination, but smaller PV area:

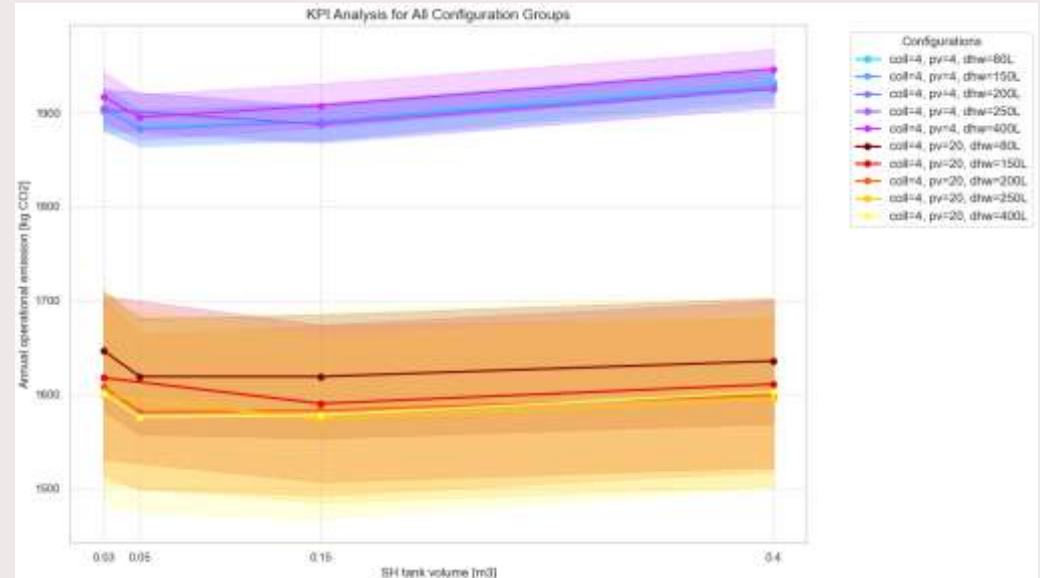
- Since controls are based on excess PV generation, a small PV system does not show any difference with different control strategies



Observation 2

A very large space heating tank does not help in reducing emissions

- Due to the absence of stratification in the tank, it needs to be charged rather soon
- And due to its large thermal inertia, it demands high energy to meet desired thresholds

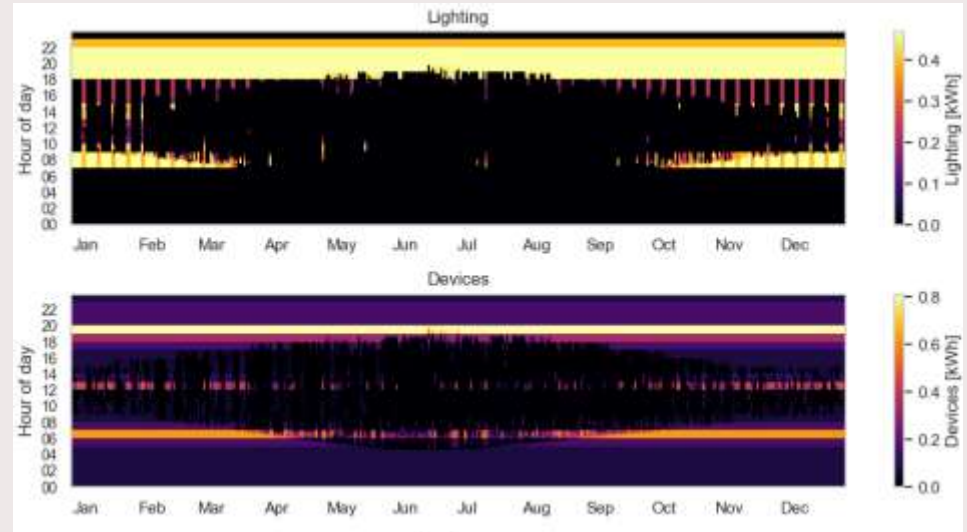


Observation 3

Even with large collectors and appropriate tank sizing, annual operational emissions cannot be reduced below 1300 kg CO₂

There is still quite some demand during the night time from lighting and devices, and even the heat pump on several days

Having an electric battery will help reduce the emissions even further



Thank you